WAGER: An Exact Decomposition of Model-Pair Score Gains on Benchmarks with Strong Label Priors

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Abstract

A scene graph generation method reports a ten-point gain in mean recall. What did it buy? On benchmarks whose labels are partly determined by a feature both compared models observe — the object-class pair in relation prediction, the class frequency in long-tailed recognition — such a gain mixes two improvements: the new model may fit the label frequencies *within* groups of that feature more closely, which a frequency table also achieves, or match its predictions to the individual cases they were made for. Mean recall, zero-shot recall and significance tests leave the two mixed. We decompose the gain exactly: score each model's prediction for one test case against the label of a *different* case in the same group: averaging over relabellings gives the *transported gain*, and what remains is a *within-group covariance gain*. The identity is exact in finite samples, needs no fitted baseline, held-out fold or tuning parameter, and costs $O(NK)$ on cached predictions. The remainder is an order-two U-statistic, holds for every Bregman score, and admits cluster-robust intervals. On two released MOTIFS checkpoints evaluated with their authors' code, a ten-point mean-recall improvement corresponds to a proper-score difference that is overwhelmingly transported. The unmodified estimator transfers to a frozen-CLIP variant and to long-tailed image and text classification, exposing compositions aggregate scores hide — including a model that loses on every such measure yet discriminates individual cases significantly better.

Keywords: Model evaluation, Benchmark auditing, Dataset bias, Scene graph

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generation, Long-tailed recognition, Proper scoring rules, U-statistics

1. Introduction

A scene graph generation method reports a ten-point improvement in mean recall on Visual Genome. A long-tailed classifier reports two points of accuracy on CIFAR-100-LT. What did those gains buy?

The question is not rhetorical, because on benchmarks like these a model can raise its score without becoming any better at the cases it is scored on. In relation prediction, the ordered object-class pair already predicts the relation strongly: given (man, surfboard), the label is very often riding, and a lookup table of training frequencies is famously competitive [1]. A new model can therefore improve by estimating $\Pr(Y \mid \text{class pair})$ more accurately — a real improvement, but one that no more distinguishes *these two men on surfboards* than the frequency table did, and one that evaporates the moment the relation frequencies shift. Or it can improve by telling those two cases apart: putting higher probability on the correct label for the individual image, beyond what the class pair alone implies. The reported score is the sum of the two, and the field's own metrics cannot separate them — not mean recall, not zero-shot recall, and not a significance test on the difference, because both components sit inside the same number.

The problem is general and the community knows it: VQA systems collapse under changed answer priors [2], and benchmark audits keep finding multimodal models exploiting non-visual shortcuts that inflate exactly this kind of aggregate score [3]. Reporting standards have responded with prior-aware slicings — frequency-stratified recall, zero-shot recall, rarity-stratified accuracy (Section 2.4) — but every one of them describes a *single* model. None attributes the gain between two submitted systems.

The customary audit builds a prior-only reference: fit $\widehat{\Pr}(Y \mid \phi)$ from training data, score it, and see how far the full models beat it [1]. As a diagnostic for one model that is useful; as an answer to the pairwise question it is indirect and noisy. The reference has to be estimated, it needs a smoothing rule for rare cells and a model family that may match neither compared system, and decomposing two models separately then subtracting gives no uncertainty for the difference — the quantity actually in dispute.

What this paper does.. We decompose the score difference itself, exactly, with no fitted reference at all. The idea takes one sentence. Both models have already emitted a full probability vector for every test case, so each model's score can be recomputed against *any* label, not only the one that was observed. Take two test cases in the same group — sharing the same value of a declared grouping variable ϕ, such as the object-class pair — and score each case's predictions against the *other* case's label. That relabelling preserves the group and its label frequencies while destroying the correspondence between a prediction and the case it was made for. Whatever score advantage survives is an advantage the new model holds over the group as a whole; whatever disappears was case-specific. Averaging within groups turns this into an identity,

$$\text{total score gain} \;=\; \underbrace{\text{transported gain}}_{\text{survives relabelling}} \;+\; \underbrace{\text{within-group covariance gain}}_{\text{destroyed by relabelling}},$$

exact in the observed sample, for any proper score, with no approximation, no held-out fold and no tuning parameter, at a cost of $O(NK)$ for N cases and K labels. Figure 1 shows the relabelling on two real Visual Genome cases. Supplementary Section S1 works the whole construction through on four data points, in arithmetic a reader can check by hand, and uses the same four points to exhibit a model-selection decision the split gets right and the aggregate score gets wrong: a candidate whose entire gain survives relabelling becomes *worse* than the model it replaces once the deployment label frequencies shift, while one whose gain does not survive is untouched.

What the second component is. The component destroyed by relabelling is not new, and naming it correctly predicts how it behaves. Written out, it equals the covariance, inside each group, between how much the new model moved its pre-

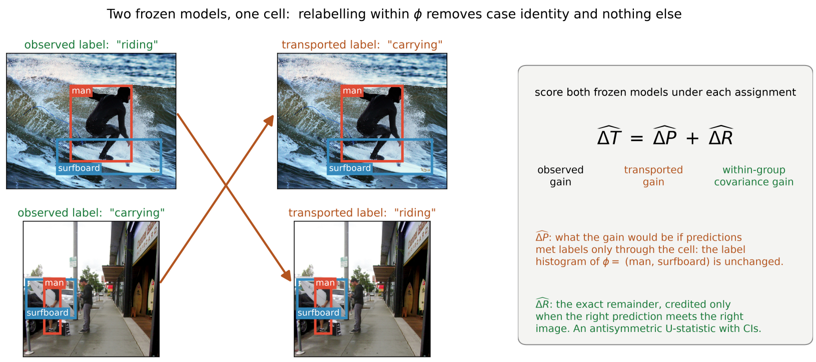

Figure 1: The construction, on two real VG150 test relations. Both share the group $\phi = (\mathsf{man}, \mathsf{surfboard})$ — the ordered object-class pair that already predicts the relation — yet carry different labels. Exchanging their labels (arrows) leaves the group and its label frequencies intact while breaking the link between each prediction and the image it was made for. Scoring two frozen models before and after the exchange splits their score difference exactly into a transported gain and a within-group covariance gain. Nothing about the construction is specific to relation prediction: it needs the two probability vectors, the labels, and the group identifiers.

dicted probability for a label and whether that label occurred (Theorem 1) — the object isolated by the covariance rearrangement of a proper score going back to Yates [4], so we call it the *within-group covariance gain*. It is close to, but not the same as, the *resolution* term of the classical reliability–resolution partition [5, 6, 7], and the difference matters in practice: classical resolution depends on a forecast only through the partition its level sets generate, so it survives any monotone rescaling of forecast values, whereas a covariance between those values does not. The two coincide exactly when both models are calibrated within groups. Section 4.1 measures the gap, and it is large enough that every comparison we report is between confidence-matched models — released checkpoints, which routinely differ sharply in calibration, cannot be audited on raw outputs.

We call the estimator WAGER; the acronym is a label, not a claim, and we do not expand it. Its two components are the *transported gain* ΔP — named for the operation that produces it, not for an interpretation of it — and the within-group covariance gain ΔR.

1.1. Contributions

1. **An audit of a released debiasing method.** On the two publicly released MOTIFS checkpoints of Tang et al. [8], evaluated on the canonical VG150

PredCls split with the original authors' code, TDE's ten-point mean-recall improvement corresponds to a proper-score difference that is overwhelmingly transported, with a within-group covariance remainder at most a small fraction of it (Section 4.2).

2. **A drop-in tool for any closed-set benchmark.** The estimator needs four aligned arrays — two $N \times K$ probability matrices, the labels, the group identifiers — runs in $O(NK)$ on cached predictions with no refitting, no held-out fold and no tuning parameter, and returns both components with cluster-robust intervals in one pass. We apply it unchanged to relation prediction, a frozen-CLIP variant, and long-tailed image and text classification; only ϕ changes. Code, cached outputs and a verification script are released.

3. **An exact identity, with inference.** For any score the gain splits exactly, in population and finite sample, into a transported gain and a remainder that is an order-two U-statistic; for every Bregman score that remainder is a within-group covariance between the models' predicted-probability difference and the label indicator, and a change constant within groups contributes exactly zero. The influence function of the difference gives image-cluster-robust intervals in one pass, the estimator is asymptotically normal with a consistent sandwich variance, and at a trivial ϕ it reduces to a clustered Diebold–Mariano test. Three further exact results remove guesswork about design choices: the in-sample plug-in is attenuated by exactly $(n_c - 1)/n_c$, merging groups shifts the estimand by an explicit between-group covariance of either sign, and the transported component splits into a mean-forecast-fit term minus a dispersion term — which is why we do not call it prior fit.

4. **Validation, and honest limits.** Simulations establish interval coverage under image clustering and delimit what the components do *not* separate: prior-only improvements stay out of the covariance channel, unrecorded within-group shortcuts enter it, and monotone recalibration moves score between the channels, which is why confidence matching is part of the pro-

cedure. Equally fine, fully identified partitions can also disagree without limit, so declaring ϕ is a modelling decision.

1.2. Terminology

Supplementary Table S3 in Supplementary Section S13 fixes the paper's vocabulary against standard counterparts. Two entries carry a caveat rather than a synonym, because no standard term matches exactly: ΔP is named for the operation that produces it, and ΔR is a covariance rearrangement's term rather than the classical resolution it is often mistaken for. We use "group" and "cell" interchangeably.

1.3. Scope, and what is not claimed

The *identity* needs four aligned arrays: two $N \times K$ probability matrices, the labels, and the cell identifiers. Any closed-set probabilistic prediction task supplies them; free-form or generative evaluation, where no K-way vector exists, does not. The *procedure* we recommend needs cluster identifiers for the intervals and, whenever the models differ visibly in confidence, a temperature fitted per model on held-out clusters (Section 4.1).

Four limits are worth stating before any result, and Section 5 returns to each. Everything is conditional on the declared ϕ: a positive ΔR says the new model predicts individual cases better *among cases sharing that feature*, and nothing about causality or robustness. Second, ΔR credits any signal varying within a cell, including an unrecorded shortcut; Supplementary Proposition S2 bounds that exposure rather than removing it, and for our own Visual Genome pair a covariate correlated at 0.061 would account for the entire reported gain (Supplementary Section S10). Third, and this is the limitation we would most want carried away, ΔR is *not invariant to recalibration*: rescaling a fixed model's confidence changes it while changing no ranking, no top-1 decision and no information, and in simulation a pure recalibration moves it by -0.488, tens of times any real gain reported here. Every result below is for confidence-matched pairs. Fourth, a transported gain is not illegitimate — fitting the deployment label distribution better is real skill when that distribution is the one that matters — and it is also not purely prior fit (Supplementary Proposition S3).

Organization.. Supplementary Section S1 works a four-point example and uses it to exhibit a decision the split changes. Section 2 places the construction in the literature on proper-score decomposition and forecast comparison. Section 3 develops the estimator and its theory. Section 4 validates it and audits two released checkpoints. Sections 5 and 6 discuss limits and conclude. Proofs are in Supplementary Section S6; three further empirical studies are in Supplementary Section S15.

2. Related work

The construction sits between two literatures that rarely meet: the decomposition of proper scores, which supplies the object being estimated, and the comparison of forecasters, which supplies the inferential target. We take those first, then the applied literature that motivates the problem.

2.1. Relation to classical proper-score decompositions

Conditional on a chosen grouping variable, a proper score splits into a calibration (reliability) term and a resolution term [5, 6], made conditioning-set explicit by Bröcker [7], whose treatment already covers general strictly proper scores and multicategory outcomes.

Our ΔR is related to that resolution term but is not equal to it, and the distinction matters before anything is built on it. The representation of a resolution-like quantity as a *covariance* between forecast and outcome indicator is due to Yates [4], whose covariance decomposition of the Brier score is the direct ancestor of Eq. (7). The classical resolution term of DeGroot and Fienberg [6] is a different object: it depends on a forecast only through the partition generated by its level sets, so it is invariant under any strictly monotone relabelling of forecast values. A covariance between forecast values and outcomes has no such invariance. The two coincide — ΔR_c equals twice a difference of classical resolution terms — exactly when both models are calibrated within each cell of ϕ, and otherwise differ by a reliability–resolution cross term that a monotone recalibration moves. Section 4.1 measures the gap: a pure recalibration of one model, which cannot change either model's classical resolution at all, moves $\widehat{\Delta R}$ by -0.488 and reverses the sign verdict of the audit in Section 4.2. We therefore name ΔR for the covariance it is, and adopt confidence matching as procedure rather than caveat.

What the classical decomposition does not supply is inference for a pairwise contrast. The point estimates do combine — both models share the ϕ-partition, so the same $(n_c - 1)/n_c$ attenuation applies to each, and subtracting two naive plug-ins recovers exactly the biased contrast — but inference does not. Variance estimators for the single-model decomposition [9] target one resolution term in isolation and supply no covariance between it and a second model's estimate from the same data. Rebuilding that from the joint law is routine rather than open [10, 11]. What is new is not that a contrast can be given a variance, but that it admits an *exact finite-sample decomposition* whose remainder is an antisymmetric U-statistic with a one-pass influence function, so the split and its uncertainty come from one construction.

2.2. *Comparing two forecasters*

Inference on a paired score differential is the classical territory of comparative forecast evaluation: the Diebold–Mariano test [12] and conditional predictive ability tests [13] ask whether one forecaster beats another, optionally conditional on covariates. That is the closest existing relative of a gain evaluated conditional on ϕ, and Corollary 1 makes the relation exact: at the trivial one-cell partition our total-gain statistic and its interval *are* a clustered Diebold–Mariano test, corrected for cluster rather than serial dependence. The decomposition is therefore best read as everything those tests already deliver plus a split that neither provides. A structurally close precedent lies outside forecast evaluation: DeLong et al. [11] compare two correlated ROC curves' areas through a generalized U-statistic covariance estimator — the same construction applied to an unconditional ranking statistic rather than a declared-cell gain.

Evaluating calibration within cells of a declared grouping, uniformly over many candidate groupings, is the subject of multicalibration [14], the formal treatment of the multiplicity question the choice of ϕ raises (Section 5).

2.3. Transport, permutation, and U-statistics

Strictly proper scoring rules reward honest predictive distributions [15], and permutation methods have long been used to assess classifier performance [16]. Conditional permutation tests generally estimate or assume a conditional distribution when the conditioning variables are continuous or high-dimensional. The setting here is different in a way that buys exactness: a benchmark supplies a *discrete* grouping variable, and the target is not generic conditional independence but one specific model-to-model score gain, so within-cell transport needs no nuisance law at all. The remainder is a U-statistic of order two in the sense of Hoeffding [17], and its Hájek projection is what supplies the influence function of Section 3.9. The new element is the gain-contrast transport identity and the attribution it licenses, not permutation or U-statistic theory by itself.

Two further connections are worth naming. The transported component is what label-shift procedures manipulate post hoc: re-weighting a classifier's outputs towards a new label distribution [18, 19] can add or remove transported gain without touching case-level evidence, which Section 4.4 exploits as a falsification test, and which is why Supplementary Section S1's model-selection reversal is a label-shift argument. And what an unrecorded grouping variable can hide is the subject of omitted-variable sensitivity analysis [20]; Supplementary Proposition S2 adapts that style of argument to bound how far an unrecorded covariate could move ΔR.

Usable-information analyses measure information accessible to a chosen predictor family; their central objects are model- or family-level quantities, whereas ours is narrower and operational, attributing the score difference between two concrete frozen models. A positive ΔR establishes nothing about causal or compositional structure.

2.4. Prior-aware evaluation in vision

Dataset bias has long complicated comparisons between recognition systems [21]. Scene graph generation was posed as structured prediction over a whole image and solved by iterative message passing between object and relation nodes [22]; against that formulation a frequency predictor of $p(\text{predicate} \mid \text{subject}, \text{object})$ is surprisingly competitive [1]; mean recall and causal interventions were proposed to reduce that dependence [8], though metric pathologies remain [23] and the same diagnosis recurs in open-vocabulary and knowledge-enhanced work [24]. In visual question answering, GQA-OOD stratifies by answer rarity within question groups [25] and VQA-CP changes language priors between train and test [2]; the shortcut is still targeted directly [26], and benchmark-level audits of exploitable non-visual shortcuts have widened to multimodal evaluation generally [3]. In long-tailed recognition a classifier can raise aggregate accuracy by fitting training class frequencies more precisely without discriminating better within a class [27]; the canonical remedies rebalance the objective or the classifier [28, 29, 30], with a continuing stream of variants [31].

Reporting standards have responded to the shortcut: mean recall alongside recall, often broken down by predicate frequency [8, 23], and zero-shot recall [32] checking whether a gain survives on triplets absent from training. These are effective prior-aware slicings, splits and stress tests of a *single* model's performance. What none provides is what we target: an exact decomposition, with uncertainty, of the score difference between two fixed systems. A stratified or zero-shot metric can show *that* a gain is uneven across the prior; it does not attribute the pairwise gain itself, on the full evaluation set, with a finite-sample identity and inference. The remedies, equally, intervene on how a model is trained rather than on how a trained pair is compared.

3. The estimator and its properties

This section states the construction of Section 1 formally and establishes its properties. The order is: notation and the object being decomposed (Section 3.1);

the population identity and the covariance form of its remainder (Section 3.2), which Section 3.4 shows is not specific to the quadratic score; what happens when the grouping variable is made coarser (Section 3.5) or is missing a relevant covariate altogether (Supplementary Section S5.1); the finite-sample identity and its $O(NK)$ computation (Sections 3.6–3.7); why the counterfactual leaves each case out rather than fitting the group's own frequencies (Section 3.8); and inference, both finite-sample and asymptotic (Sections 3.9–3.10). Proofs are in Supplementary Section S6. A reader who wants only what is needed to use the estimator can read Sections 3.1, 3.2, 3.6 and 3.9 and skip the rest.

3.1. Setup: what is being decomposed

Let $q_0(\cdot \mid x)$ and $q_1(\cdot \mid x)$ be the predictive distributions of an old and a new frozen model over K labels. Let $S(q, y)$ be a bounded proper score, oriented so that larger is better, and let $C = \phi(X)$ be a declared discrete grouping variable. In the relation-prediction example of Section 1, C is the ordered subject–object class pair; in a long-tailed classification benchmark it is a coarse superclass. We call a value of C a *cell* and require it to be declared before the audit is run. Define the instance-specific *gain vector*

$$H_x(y) = S(q_1(\cdot \mid x), y) - S(q_0(\cdot \mid x), y), \qquad y = 1, \ldots, K. \tag{1}$$

Unlike the realized gain $H_X(Y)$, this vector can be evaluated at a counterfactual label without rerunning either model.

Our default is the quadratic score

$$S_{\mathrm{B}}(q, y) = 2q_y - \|q\|_2^2, \tag{2}$$

which differs from negative Brier loss only by a label-only constant. It is strictly proper and bounded, avoids probability clipping, and uses the entire predictive distribution. A floored log score is a sensitivity analysis.

3.2. Within-cell counterfactual transport

For a cell c, define three population quantities:

$$\Delta T_c = \mathbb{E}[H_X(Y) \mid C = c], \tag{3}$$

$$\Delta P_c = \mathbb{E}_{H|c}\, \mathbb{E}_{Y'|c}[H_X(Y')], \tag{4}$$

$$\Delta R_c = \Delta T_c - \Delta P_c. \tag{5}$$

Here Y' is an independent label drawn from the same cell. Thus ΔT_c is the observed score gain, ΔP_c is the gain retained after breaking the instance-specific prediction–label assignment while preserving the cell, and ΔR_c is the covariance gain lost by that transport.

Theorem 1 (Transport and covariance identities)**.** *For any integrable score contrast H,*

$$\Delta T_c = \Delta P_c + \Delta R_c, \qquad \Delta R_c = \sum_{y=1}^{K} \mathrm{Cov}(H_X(y), \mathbb{1}\{Y = y\} \mid C = c)\,. \tag{6}$$

For the quadratic score, writing $\Delta q_y(X) = q_1(y \mid X) - q_0(y \mid X)$*,*

$$\Delta R_c = 2 \sum_{y=1}^{K} \mathrm{Cov}\Big(\Delta q_y(X), \mathbb{1}\{Y = y\} \mid C = c\Big)\,. \tag{7}$$

Consequently, if the gain vector is a function of c alone, then $\Delta R_c = 0$*.*

Equation (7) is the operational meaning of the construction: the new model receives covariance credit only to the extent that its probability change aligns more strongly with the label of the correct case within the same cell. Cell-constant score improvements, including a better fit to the cell's label histogram, remain in ΔP. A nonzero covariance does not assert causal or compositional understanding; unrecorded within-cell shortcuts contribute to it too, so ϕ must be declared and stress-tested.

3.3. *What the transported gain measures*

The surviving component is easy to misname. It is tempting to call ΔP a prior-fit gain, since transport replaces a case's label by an independent draw from its cell; earlier drafts of this paper did. Supplementary Proposition S3 in Supplementary Section S14 shows the name is wrong and says what the quantity is instead: under the quadratic score ΔP_c equals the improvement in how closely the new model's *mean* within-cell forecast sits to the cell's label frequencies, minus the increase in the within-cell dispersion of its forecasts. Only the first term is prior fit. The second rewards the new model for making less dispersed predictions at a fixed mean forecast, so ΔP can absorb an entire observed gain with both models' mean forecasts sitting exactly on the cell frequencies — a four-point counterexample in that appendix does just that. This is the reliability-and-sharpness channel of the classical partition, not a prior-fitting channel, which is why we name it for the operation that produces it; it is also why confidence matching moves $\widehat{\Delta P}$ so sharply, since a temperature change is almost purely a change in dispersion.

3.4. *Generalization: the Bregman-score family*

Nothing in the proof of Theorem 1 uses properness: the identity and the covariance sum follow from the definitions alone, for any score. What is specific to the quadratic score is the further reduction to Eq. (7), in which the part of $H_x(y)$ not depending on y integrates out because $\sum_y \mathbb{1}\{Y = y\} = 1$ almost surely. The same cancellation occurs for every strictly proper score generated by a Bregman divergence, so Eq. (7) and the log-score identity used as a robustness check in Section 4.2 are two instances of one theorem. For a strictly convex, differentiable ψ on the simplex, the Bregman score is $S_\psi(q, y) = \psi(q) + \langle \nabla\psi(q), e_y - q \rangle$; taking $\psi(q) = \|q\|_2^2$ recovers the quadratic score and $\psi(q) = \sum_k q_k \log q_k$ the log score [15].

Supplementary Section S2 states the resulting identity and proves it: for a Bregman score generated by ψ, the remainder equals the same within-cell covariance sum with the probability difference replaced by the contrast of the y-th gradient coordinate of ψ. Two named cases follow. For $\psi(q) = \|q\|_2^2$ the gradient contrast is $2\Delta q_y(x)$ and the identity is Eq. (7); for the negative entropy it is the log-likelihood-ratio contrast $\log q_1(y \mid x) - \log q_0(y \mid x)$, so the log-score check of Section 4.2 carries the same exact covariance meaning rather than being merely a sensitivity analysis. The implemented log score floors each probability at $\epsilon = 10^{-6}$ to avoid $-\infty$ on an unseen label; the identity is exact for the unfloored score and the floored version is a check on it, negligibly different whenever no relevant probability is within ϵ of zero.

3.5. Coarsening the cell, and an unrecorded covariate

Two questions about the declared ϕ have exact answers, both stated and proved in Supplementary Section S5. First, what happens when ϕ is made coarser: merging cells changes the estimand by an explicit between-cell covariance term, taken over the finer cells nested inside each coarse one, between each fine cell's typical gain at a label and that label's frequency there. The term is not sign definite, so a coarser grouping can inflate or deflate the credited covariance gain rather than leaving it alone, and it vanishes only when one of the two quantities is constant within each coarse cell. This is why the finest defensible ϕ is the conservative default, and Section 4.5 reports the empirically inflating case.

Second, read in the other direction, the same identity says what happens when a relevant covariate is *never* declared — the estimator's main scope limit. Writing Z for the unrecorded variable and (ϕ, Z) for the fully adjusted cell, the bias of using ϕ alone is exactly that between-cell covariance term with (ϕ, Z) as the fine partition. It is exact but needs Z's joint law with the gain and the label, so Supplementary Section S5.1 replaces it with a bound requiring only a single elicited correlation $\bar{\rho}$ between an unrecorded variable's effect on the within-cell gain and its effect on the within-cell label distribution, in the spirit of the robustness value that Cinelli and Hazlett [33] elicit for regression confounding: the bias is at most $\bar{\rho}$ times a product of two dispersion factors, with a data-free form $\bar{\rho} M \sqrt{K-1}$ needing only the label cardinality and the score bound. Because $\bar{\rho}$ is a correlation rather than a domain-specific odds ratio it can be elicited the same way across benchmarks. Both forms are evaluated on our own estimates there, and the sharp one is the less comfortable: an unrecorded covariate correlated at 0.061 would account for the entire covariance gain we report in Section 4.3.

3.6. Exact finite-sample decomposition

Consider cell c with $n_c \geq 2$ examples. Write $H_i(y) = H_{x_i}(y)$. WAGER computes

$$\widehat{\Delta T}_c = \frac{1}{n_c} \sum_{i \in c} H_i(y_i), \tag{8}$$

$$\widehat{\Delta P}_c = \frac{1}{n_c(n_c - 1)} \sum_{i \neq j \in c} H_i(y_j), \tag{9}$$

$$\widehat{\Delta R}_c = \binom{n_c}{2}^{-1} \sum_{i < j \in c} A_{ij}, \tag{10}$$

where the antisymmetric kernel is

$$A_{ij} = \frac{1}{2}\{H_i(y_i) + H_j(y_j) - H_i(y_j) - H_j(y_i)\}. \tag{11}$$

For every cell with $n_c \geq 2$ these satisfy $\widehat{\Delta T}_c = \widehat{\Delta P}_c + \widehat{\Delta R}_c$ exactly, with no distributional assumption, and when the cell's cases are drawn i.i.d. from the cell population $\widehat{\Delta R}_c$ is an unbiased order-two U-statistic for ΔR_c; Supplementary Section S3 states and proves both, and notes that independence rather than exchangeability alone is what unbiasedness requires, since the transported quantity pairs a case with a label drawn independently from the same cell.

Dataset-level quantities weight cells by their identified counts, $\widehat{\Delta Z} = N_*^{-1} \sum_{c:n_c \geq 2} n_c \widehat{\Delta Z}_c$ for $Z \in \{T, P, R\}$. Singleton cells cannot identify a within-cell counterfactual, so we exclude and report them rather than inventing a smoothed estimate. If $\widehat{\Delta T} > 0$ the descriptive share $\widehat{\Delta R}/\widehat{\Delta T}$ may exceed one, when covariance improves while the transported channel deteriorates, and it is undefined for a nonpositive total; we therefore report $\widehat{\Delta R}$ as the primary result rather than a thresholded ratio.

3.7. Linear-time computation

Naively evaluating Eq. (9) is quadratic in cell size. Let m_{cy} be the count of label y in cell c. For example $i \in c$, its transported score is

$$P_i = \frac{\sum_{y=1}^{K} m_{cy} H_i(y) - H_i(y_i)}{n_c - 1}. \tag{12}$$

Averaging P_i yields Eq. (9); averaging $H_i(y_i) - P_i$ yields Eq. (10). Thus the entire estimator costs $O(NK)$ time and $O(NK)$ storage for cached prediction vectors. No model fitting, projection fold, hierarchy, or smoothing hyperparameter appears.

3.8. Why leave-one-out transport, not a fitted-in-sample prior

Equation (12) excludes example i's own label from its transported score. It is natural to ask what is lost by the simpler alternative of fitting the cell's empirical label frequency on all n_c examples, including i, and using that fitted frequency as the counterfactual, i.e. treating $\hat{p}_c(y) = m_{cy}/n_c$ as a prior model and setting $\tilde{P}_i = \sum_y \hat{p}_c(y) H_i(y) = n_c^{-1} \sum_y m_{cy} H_i(y)$. This is the natural plug-in estimator, and it is what a frequency-collapsed projection baseline computes when no separate evaluation fold is withheld from fitting.

Proposition 1 (Exact attenuation of the in-sample plug-in)**.** *For every cell with $n_c \geq 2$ and every $i \in c$, writing $\hat{P}_i$ for Eq.* (12) *and $\hat{R}_i = H_i(y_i) - \hat{P}_i$,*

$$\tilde{P}_i = \hat{P}_i + \frac{1}{n_c} \hat{R}_i, \qquad \tilde{R}_i := H_i(y_i) - \tilde{P}_i = \frac{n_c - 1}{n_c} \hat{R}_i. \tag{13}$$

Consequently the in-sample plug-in's dataset-level statistic obeys $\widetilde{\Delta R} = N_*^{-1} \sum_{c:n_c \geq 2} (n_c - 1) \widehat{\Delta R}_c$, a multiplicative shrinkage of every cell's leave-one-out covariance gain by the factor $(n_c - 1)/n_c$.

The shrinkage is deterministic, not a variance statement: whenever $\widehat{\Delta R}_c \neq 0$ the in-sample plug-in is biased for it at every finite n_c, worst in the smallest identified cells (50% at $n_c = 2$). The standard remedy for such self-prediction bias is a projection/evaluation split, which restores unbiasedness by discarding examples from both stages and introduces a split-fraction hyperparameter and typically higher variance. Leave-one-out transport removes the bias exactly, at the full sample, for every cell with $n_c \geq 2$, with no fitted nuisance distribution at all.

3.9. Inference

For point estimation we use all unordered within-cell pairs. For uncertainty, define $\bar{A}_i = (n_c-1)^{-1} \sum_{j \neq i} A_{ij}$ and $\widehat{\Delta R}_c = n_c^{-1} \sum_i \bar{A}_i$. The estimated Hájek influence contribution for random cell membership is

$$\widehat{\psi}_i = 2\bar{A}_i - \widehat{\Delta R}_c - \widehat{\Delta R}. \tag{14}$$

We sum $\widehat{\psi}_i$ within image and use the resulting one-way sandwich variance, so multiple relations from the same image are not treated as independent. Total-gain influence is $H_i(y_i) - \widehat{\Delta T}$; prior-gain and share intervals follow from the exact identity and delta method.

As a complementary finite-group diagnostic we apply an independently sampled cyclic shift to the labels inside every cell, preserving cell size and label histogram, and recompute $\widehat{\Delta R}$ under B shifts to obtain a one-sided p-value; Supplementary Section S8 gives the algorithm. It is exact under the sharp null that labels are exchangeable relative to the frozen model outputs within each cell. Because relations share images, our primary real-data uncertainty statement is the image-cluster-robust interval and the randomization test is a secondary relation-level diagnostic.

3.10. Asymptotic normality

The influence function and sandwich variance behind the reported intervals are derived in Supplementary Section S7 through a Hájek-projection expansion, the standard route for order-two U-statistics [34, 35]. The following supplies the limit theorem, under conditions that hold whenever the score is bounded and no cluster dominates the sample.

Supplementary Theorem S3 in Supplementary Section S11 states it: if the score is bounded, examples fall in mutually independent clusters none of which dominates the sample, the number of eligible cells is negligible against the identified sample, and ϕ has fixed finite support whose every cell grows, then $\sqrt{N_*}(\widehat{\Delta R} - \theta)$ is asymptotically normal around the identified-subpopulation estimand and the implemented sandwich variance is consistent for its limiting variance. The last condition is an idealisation no heavy-tailed benchmark attains in sample, which is why Section 4.1's coverage simulation, not this theorem, is what licenses reading the reported intervals at finite N.

Relation to comparative forecast evaluation..

Corollary 1 (Relation to Diebold–Mariano/Giacomini–White)**.** *Under the trivial partition $\phi \equiv$ const (one cell containing every example), $\widehat{\Delta T} = N^{-1} \sum_i H_i(y_i)$ is the sample mean of the paired loss differential between the two models, and Supplementary Theorem S3 applied to $\widehat{\Delta T}$ with its image-clustered interval is the grouped, cluster-robust form of the Diebold–Mariano statistic for that differential [12]; because the conditioning information set is empty, it is equivalently a degenerate instance of the Giacomini–White conditional predictive-ability test [13].*

The corollary is immediate from the definitions: DM and GW test whether a paired loss differential's mean is zero, and $\widehat{\Delta T}$ at the trivial partition is exactly that differential's sample mean. Neither test forms $\widehat{\Delta P}$ or $\widehat{\Delta R}$; both target ΔT and stop. One qualification on how far the reading travels: the corollary is stated at the trivial partition, whereas the totals reported at a nontrivial ϕ are N_*-weighted means over the *identified* subsample. Because ΔT does not depend on ϕ except through that exclusion, each reported $\widehat{\Delta T}$ is a clustered DM statistic for the paired differential on its identified subsample, which at the coverages here (99.0% on VG150, 98.9% in Section 4.2) differs from the full-sample statistic only in the excluded singletons.

4. Experiments

Section 4.1 validates the instrument under controlled alternatives where the truth is known by construction, and establishes what it fails to separate. Section 4.2 is the paper's main result: an audit of two publicly released checkpoints of a widely cited debiasing method, chosen so that the audited models are not ours and the claimed improvement is one the field already reports. Sections 4.3–4.4 then return to relation prediction with predictors built so that their information content is controlled, which is the complementary evidence a released checkpoint cannot give: there the ground truth about what each model can see is known, so the decomposition can be checked against it. Two further studies — long-tailed image classification on CIFAR-100-LT across seeds and imbalance ratios, and long-tailed text classification on 20-Newsgroups-LT — apply the identical estimator in other data modalities and are reported in Supplementary Section S15, together with the granularity, score and minimum-cell-size sensitivity analyses. All confidence intervals are 95%.

4.1. Controlled validation

We generate 24 cells over five labels, drawing each cell's label distribution from a Dirichlet law and taking the old model to be exactly the cell prior. The new model interpolates towards an oracle, $q_1 = (1 - \beta)q_0 + \beta q_{\text{oracle}}$ with q_{oracle} placing 0.98 on the realized label; forty repetitions run at each $\beta \in \{0, .1, \ldots, .8\}$. The estimated covariance gain grows linearly and monotonically with the injected signal. Type-I behaviour is assessed by perturbing the prior predictions with mean-zero noise and reassigning the perturbed vectors within each cell, which guarantees no systematic prediction–label alignment survives: over 200 such datasets the 99-shift randomization test rejects in 0.030 of runs at level 0.05, with a mean estimate of -8.97×10^{-5} and approximately uniform null p-values.

Coverage of the primary interval.. The paper's primary uncertainty statement is the image-clustered Hájek interval, so we validate it in the regime that stresses it: relations grouped into images (one to about thirty each) whose oracle weight shares a latent per-image effect, with Zipf-distributed cell sizes in which most identified cells hold exactly two examples. Against the design expectation from 4,000 replications, the cluster-robust interval covers in 94.0% of 500 runs while an interval treating relations as independent covers in only 88.6% — quantifying both the mild finite-sample undercoverage and the necessity of clustering.

Discriminant validity. Three designs test that the channels separate what they claim to. A *calibration-only* alternative, where the new model is a temperature-softened copy of an overconfident informative model so the pair carries identical case information, is credited raw with a large negative covariance gain (mean -0.488), which the protocol below reduces to -0.0003 (sd 0.0017 over 200 runs). A *prior-only* improvement, where the new model corrects a heteroscedastically noisy estimate of the cell prior, yields a covariance gain centred at zero (0.0002, sd 0.0035) with a clearly positive transported channel. And an *unrecorded within-cell shortcut* — a binary feature varying inside each cell that shifts the label distribution and is used by the new model — is credited to covariance (+0.041, sd 0.004): $\widehat{\Delta R}$ measures all within-cell predictive signal relative to the declared ϕ, shortcut or otherwise.

The calibration-matched protocol.. The first of those designs identifies a limitation serious enough to be part of the procedure rather than a caveat about it: because ΔR is a covariance between probability movements and labels (Theorem 1), rescaling a fixed model's confidence moves score between the two components while changing no ranking, no decision and no case-level information. We therefore adopt the following control wherever the compared models differ visibly in confidence. Split the evaluation units — images, where examples are clustered — at random into halves. On the first half, fit each model a single temperature T by maximum held-out likelihood on $q^{1/T}$ renormalized. On the second half, apply each model's own fitted temperature and run the decomposition there. The protocol costs nothing but a data split, it uses no label information from the audited half, and it is what the simulation above shows removes the confidence-scaling channel almost exactly.

How many splits each study averages over differs, and the difference is worth stating rather than glossing. The long-tailed classification study of Supplementary Section S15.5 averages over twenty random splits. The audit of Section 4.2 and the matched rows of Section 4.4 each use a single split at a fixed seed. For those two the split is a source of variability the reported interval does not include, and it is not negligible: the calibration-only arm above has a between-split standard deviation of 0.001725, exceeding the half-width of the audit's own reported interval (0.00115). The audit's conclusion does not turn on this — its estimate is indistinguishable from zero and stays so under any split — but a reader comparing interval widths across our tables should know that two of the three matched studies report within-split uncertainty only. It does not make ΔR invariant to recalibration in general; it fixes the confidence scale at which the comparison is made, which is what a comparison of two *fixed* models needs.

4.2. Auditing a published debiasing method

The question the decomposition exists to answer concerns systems other people release and the field already compares, so the primary application is to a published pair. Tang et al. [8] release a causal MOTIFS PredCls checkpoint that yields two models from one set of weights: the biased baseline (MOTIFS-SUM) and its Total Direct Effect counterpart (MOTIFS-TDE), which subtracts a context-only counterfactual at inference. TDE is among the most cited debiasing methods in scene graph generation and is reported through mean recall, so it is exactly the kind of claimed improvement whose composition is at issue.

We evaluate both variants with the authors' own code on the canonical VG150 test split, which fixes the vocabulary, split and relation set and makes the two prediction sets aligned relation-for-relation. Our reproduction matches the published behaviour: the baseline reaches $R@50 = 0.6612$ with $mR@50 = 0.1459$, and TDE trades the former for the latter, reaching $mR@50 = 0.2476$ and raising zero-shot recall to $zR@50 = 0.1432$. Top-1 predicate accuracy falls from 0.6981 to

0.5577. The audit uses 183,639 relations over 26,446 images and 7,127 class-pair cells, of which 98.9% are identified.

What is audited. The calibration-matched rows do not decompose the checkpoints as released; they decompose the two models after each is rescaled by its own temperature ($T^* = 2.50$ for the baseline, 0.92 for TDE) fitted on a held-out half of the images. That is the comparison we argue is meaningful, because a proper score punishes miscalibration and these models differ sharply in it. The released pair, scored as it ships, gives a *significant* covariance loss (-0.01355, interval excluding zero). We report both.

Result.. Read raw, TDE loses 0.11651 of quadratic score, of which 0.10296 sits in the transported channel and a small but significant 0.01355 in covariance. That reading does not survive the control. Once temperatures are fitted on held-out images and the audit runs on the other half, the quadratic-score covariance channel is indistinguishable from zero: $\widehat{\Delta R} = -0.00006$ with interval $[-0.00120, +0.00109]$ and randomization $p = .555$. Essentially the entire proper-score difference, -0.18258 of -0.18264, is transported.

The log-score row of the same calibrated comparison does not repeat the null: it returns $+0.04396$, CI $[+0.04079, +0.04714]$, excluding zero. The two scores therefore disagree on whether any real covariance change survives confidence matching, while agreeing that whichever value is right is far smaller than the calibrated transported shift (-0.18258 under the quadratic score, -0.59157 under the log score). We do not adjudicate: nothing in Section 3.1 privileges one proper score, and Supplementary Theorem S1 is why this is not a defect to argue away — each Bregman generator defines its own exact estimand, so two scores disagreeing are answering two well-posed questions, and which one an evaluator wants is a declaration. The claim the audit supports is what both scores agree on: TDE's proper-score improvement over the baseline is overwhelmingly transported *at the calibration-matched, class-pair configuration*, with a covariance remainder at most a small fraction of the total and possibly exactly zero.

Table 1: WAGER audit of released checkpoints on the canonical VG150 PredCls split: MOTIFS-TDE against the MOTIFS-SUM baseline it debiases, both evaluated with the original authors' code. Raw rows use all identified relations; the calibration-matched rows fit each model's temperature on a held-out half of the images and audit the disjoint half. CI is image-cluster robust on $\widehat{\Delta R}$.

Regime	Score	Cell ϕ	$\widehat{\Delta T}$	$\widehat{\Delta P}$	$\widehat{\Delta R}$ (95% CI)
raw	quadratic	class pair	−0.11651	−0.10296	−0.01355 [−0.01504, −0.01206]
raw	log	class pair	+0.03277	+0.15897	−0.12619 [−0.13093, −0.12145]
raw	quadratic	subject	−0.11536	+0.10175	−0.21711
calibration-matched	quadratic	class pair	−0.18264	−0.18258	−0.00006 [−0.00120, +0.00109]
calibration-matched	log	class pair	−0.54760	−0.59157	+0.04396 [+0.04079, +0.04714]

That qualifier is load-bearing. Table 1's five configurations of this one pair give covariance gains of −0.21711, −0.12619, −0.01355, −0.00006 and +0.04396, and at two of the five "overwhelmingly transported" is simply false: the raw log-score row has $|\widehat{\Delta R}| = 0.126$ against $|\widehat{\Delta P}| = 0.159$, and the subject-only row has −0.217 dominating +0.102. We report all five and argue for one on stated grounds — matched confidence, because a proper score punishes miscalibration; the finest defensible cell, because Supplementary Proposition S1 makes coarsening non-neutral.

Reading. This characterises what TDE does rather than indicting it: its construction subtracts the context-only prediction, so a method operating on the transported channel is what its authors built, and the decomposition quantifies that the intervention is almost purely of that kind. The consequence for evaluation needs care. Mean recall rose 10.2 points; the calibration-matched quadratic covariance gain is −0.00006 with an interval containing zero. Those are different units and we supply no bridge: nothing licenses saying ten points bought 0.00006 of case-level evidence, and it would be equally wrong to say the mean-recall gain *is* transported, since the transported channel on the proper score *fell*. What the audit supports is a conjunction — the headline metric moved ten points, and the proper-score difference decomposes almost entirely into the transported channel — from which a reader may conclude that the metric movement is not evidence about within-cell case-level discrimination, but may not read off a conversion rate.

4.3. Relation prediction with controlled predictors

A released checkpoint cannot tell us whether the decomposition is measuring what it claims, because the ground truth about what the model can see is unavailable. We therefore also compare five predictors on VG150 whose information content is controlled by construction: a frequency table (FREQ) that sees only the class pair, an MLP on class embeddings (MLP-CLASS), and MLPs adding box geometry (MLP-SPATIAL, MLP-SPATIAL+). Full protocol, per-pair tables and the accuracy figures are in Supplementary Section S15.1; the audit covers 227,337 of 229,605 test relations, a coverage of 99.0%, over 6,346 identified class-pair cells.

Two readings matter here. MLP-CLASS improves on FREQ by 0.03386 of quadratic score and the decomposition assigns *all* of it to the transported channel: the covariance gain is 0.00000 with a zero-width interval and randomization $p = .724$. That zero is not an empirical finding but an algebraic one — both models are functions of ϕ alone, so their gain vector is constant within each cell and Theorem 1 forces the remainder to vanish — and it is the sharpest available check that the estimator credits nothing to case-level evidence when none exists. Adding box geometry then produces a covariance gain of 0.01439 (CI [0.01373, 0.01504], $p = .002$) that no aggregate score separates out, and against MLP-CLASS the total gain of 0.00885 actually understates it, because the transported channel deteriorates by 0.00554 — the same structure as model B of Supplementary Section S1, now on real predictors.

4.4. Real pixels against a geometric proxy

None of the predictors above ever sees an image: they are functions of the class pair and, for MLP-SPATIAL(+), of box coordinates recorded in the annotation. Since box geometry is only a proxy for image content, we also compare a model that reads pixels — subject and object regions cropped from the original image and encoded by a frozen CLIP ViT-B/32 [36], concatenated with the class embeddings the class-only model already uses. CLIP is never fine-tuned, so whatever covariance the model gains reflects information its pretraining had already recovered. All three compared predictors are retrained on one fixed subsample of 100,000 training relations so that any gap between them isolates the features rather than the training-set size; the full protocol, tables and figure are in Supplementary Section S15.3.

This is the sharpest case in the paper for separating the components. By every aggregate measure the CLIP model is the worst of the three: top-1 accuracy 0.6161 against 0.6467 for the geometry model, and a quadratic score 0.04655 lower. An evaluator reading either number would discard it. The decomposition contradicts that reading on the axis that matters. Against the geometry model the CLIP model loses 0.05296 of transported gain while *gaining* 0.00641 of covariance (95% CI $[0.00514, 0.00769]$, randomization $p = .002$): its entire deficit sits in the transported channel and its within-cell discrimination is significantly better, not worse. Against the class-only baseline, pixels buy 0.01665 of covariance where box geometry buys 0.01023, with well-separated intervals. Real image content carries more case-specific signal than the geometric proxy standing in for it, and a benchmark whose headline metric rewards transported gain will nonetheless rank the pixel model last. Two falsification checks support the reading: post-hoc prior correction registers as almost pure transported movement (+0.02401 against +0.00111), and applied to the CLIP model it halves the aggregate deficit (-0.04655 to -0.02143) while the covariance advantage persists (+0.00752, and +0.00663 when both models are corrected) and survives confidence matching (+0.00467, CI $[+0.00314, +0.00621]$).

4.5. Other modalities, and sensitivity

Supplementary Section S15 reports two further studies and the sensitivity analyses. On CIFAR-100-LT, with the benchmark's coarse superclasses as ϕ and a residual backbone [37], two standard re-weighting schedules have opposite compositions, and the confidence-matched reading reverses what the raw scores suggest about both: the schedule whose aggregate score barely moves has lost within-class discrimination, while the one that improves has bought covariance rather than the prior refit its raw decomposition implies, replicating across seeds and imbalance ratios. A long-tailed split of the 20-Newsgroups corpus [38] runs unmodified and returns interpretable output, but rests on a single trained pair per schedule with no confidence-matched row, so we report it as transfer rather than corroboration. On sensitivity: substituting a floored log score, coarsening ϕ to the subject class, and raising the minimum cell size from 2 to 20 all move the estimate, and the coarsening moves it most, from 0.01439 to 0.02181 — the between-cell covariance term of Supplementary Section S5 in action. The finest defensible ϕ is therefore the conservative choice, and Section 5 explains why even that rule is weaker than it looks.

5. Discussion and limitations

5.1. What the decomposition does and does not establish

The unit of analysis is a model-pair gain rather than a model, and that is what makes the two failure modes visible. A model may estimate within-group label frequencies better without predicting any individual case better, and be credited as though it had done the second. Conversely it may discriminate cases better while losing enough prior fit to hide the improvement — Section 4.4 contains a case where the aggregate score and the accuracy both point the wrong way. The signed identity $\Delta T = \Delta P + \Delta R$ exposes both without a fitted reference forecaster.

Everything is conditional on the declared ϕ. A positive ΔR means the new model matches its probability changes to the correct case better than the old one *among cases sharing* ϕ; it proves nothing about causal inference, compositional understanding, or robustness. If an unrecorded shortcut varies inside a cell, its predictive use is counted in ΔR, and Supplementary Proposition S2 bounds that exposure rather than removing it — at a level that, for our own Visual Genome estimates, is not comfortable (Supplementary Section S10). Applications should declare ϕ before seeing results, justify it from how the benchmark was built, and report finer and coarser alternatives, because coarsening adds a between-cell covariance term of either sign (Supplementary Proposition S1) and only the finest identified partition nets out every regularity expressible in ϕ. Even that rule is weaker than it looks: Supplementary Section S12 shows two equally fine, fully identified partitions of the same four cases that disagree completely about whether an improvement is case-level or transported, so the choice of ϕ is a modelling decision and not a formality.

Two limits deserve stating flatly. First, ΔR is not invariant to monotone recalibration: it is a covariance between probability movements and labels, so sharpening or softening a fixed model shifts score between the channels while changing no ranking, no top-1 decision and no information about any case. Section 4.1 measures the effect at -0.488, tens of times any real gain reported here, which is why we audit confidence-matched pairs and why released checkpoints, which routinely differ sharply in calibration, cannot be compared on their raw outputs. A rank-based within-cell contrast would be invariant to this, at the cost of the exact additivity that is the point of the construction; Supplementary Section S12 says what that trade would involve. Second, a transported gain is not evidence of anything illegitimate. Fitting the deployment label distribution better is real skill whenever that distribution is the one that matters; it is fragile precisely under label shift, where it can be added or removed post hoc by re-weighting [18, 19], as Section 4.4 demonstrates and Supplementary Section S1 exploits. Nor is it purely a matter of prior fit (Supplementary Proposition S3). The decomposition reports where a gain lives, not whether it is worth having.

5.2. *Choosing the grouping variable, and what the estimator cannot do*

"Declare ϕ before looking" is advice, not a mechanism. The annotation prior is a property of the dataset, so we recommend that the *benchmark*, not the submitting author, fix it, which removes the incentive to report only the most favourable candidate; that only moves the discretion, since maintainers often author methods evaluated on their own benchmark. What we would defend is the weaker claim that ϕ be fixed and public *before* a leaderboard opens, by whoever does it, with every pre-declared candidate reported as our sensitivity rows are. Guarantees holding uniformly over a class of groupings are the subject of multicalibration [14], and porting that to pairwise attribution is a natural extension. For comparability we suggest a fixed block per audited pair: ϕ and its rationale, the identified coverage, both signed components with the covariance interval, and the confidence-matched and granularity rows.

Exact transport needs at least two cases per cell, so a very fine ϕ buys confound control at the cost of coverage; we report the identified fraction and claim nothing for singletons. The image-clustered interval is asymptotic, and Supplementary Section S11 justifies it in a regime no heavy-tailed benchmark attains in sample: of 6,346 eligible class-pair cells, 2,286 hold fewer than five cases, though those carry only 2.7% of identified relations. The reported 94.0% empirical coverage against a nominal 95% therefore remains the operative evidence for the interval. The randomization test is finite-group exact only under a sharper within-cell exchangeability null and treats relations rather than images as the randomized units, so it stays a secondary diagnostic.

Finally, the estimator reads probability vectors, not top-1 decisions. That is a feature — it detects useful redistribution below the argmax — but it needs released logits or probabilities, and that is the real obstacle to routine use: the audit of Section 4.2 needed nothing but two checkpoints evaluated with their authors' code, and what blocks the same workflow elsewhere is disclosure, not compute, since benchmarks overwhelmingly archive ranked predictions. Nothing in Sections 3.1–3.9 is specific to relation prediction, and the long-tailed studies test that in two other modalities. The constraint on ϕ shows its teeth there: the fine class label cannot serve as a cell, since it would leave every cell label-homogeneous and drive ΔR to zero by construction, an artefact that reads exactly like a null result.

6. Conclusion

When two models are compared on a benchmark whose labels are partly determined by a feature both of them observe, the reported gain answers a question nobody asked. It sums a closer fit to the label frequencies within groups of that feature and a closer match between predictions and the individual cases they were made for — different achievements with different consequences, since the first is skill a frequency table also supplies and that a shift in label frequencies removes, and the second is not.

This paper separates them exactly. Scoring each prediction against the label of a different case in the same group leaves the group's label frequencies intact while breaking the correspondence between prediction and case, so the gain surviving that relabelling is the transported component and the remainder is a within-group covariance gain. The identity holds in the observed sample for any proper score, with no fitted reference model, no held-out fold and no tuning parameter, at $O(NK)$ on cached predictions. The remainder is an order-two U-statistic equal, for every Bregman score, to a within-group covariance between the models' probability change and the label indicator, and it carries cluster-robust intervals, a randomization test and a limit theorem.

What it buys. Pointed at two released MOTIFS checkpoints with their authors' own code, the decomposition shows a ten-point mean-recall improvement corresponding to a proper-score difference that is overwhelmingly transported. On predictors whose information content we control it returns exactly zero covariance gain when both models are functions of the grouping variable alone, and shows a frozen-CLIP model that loses on every aggregate measure discriminating individual cases significantly better than the model it appears to lose to.

Weaknesses. Three are structural. The estimator is not invariant to recalibration, so we compare only confidence-matched pairs and released checkpoints cannot be audited on raw outputs. It credits any signal varying inside a cell, shortcuts included, and our own sensitivity bound is not reassuring: an unrecorded covariate correlated at 0.061 would account for the entire covariance gain we report on Visual Genome. And the estimand is not determined by the granularity of ϕ — two equally fine, fully identified partitions of the same four cases can disagree completely — so declaring ϕ is a modelling decision that choosing the finest defensible partition does not settle.

Future work. A rank-based within-cell contrast would be invariant to the recalibration this construction is sensitive to, at the cost of the exact additivity that is the point here; reporting both is what we would recommend to a benchmark able to choose. Approximate matching would extend the identity to continuous grouping variables, trading exactness for modelling assumptions. And porting multicalibration-style guarantees to pairwise attribution would address the choice of ϕ more convincingly than a declaration rule can. The obstacle to all of it is the one that limits the method today: benchmarks archive ranked predictions, not probabilities.

Supplementary material

Proofs of all numbered results, the influence-function derivation, implementation and reproducibility details, two further empirical studies (long-tailed image and text classification) and the granularity, score and minimum-cell-size sensitivity analyses are provided as supplementary material, whose sections, tables and figures carry an S prefix and are cited in that form throughout.

Data and code availability

All code, cached model outputs and scripts needed to reproduce every number, table and figure are publicly available at `https://github.com/vinhqdang/WAGER`: the estimator, its unit tests, every experiment driver, and the cached results each reported value is checked against by an included verification script. The two audited scene-graph checkpoints are the public release of Tang et al. [8]; Visual Genome [39] and CIFAR-100 are public benchmarks.

CRediT authorship contribution statement

Quang-Vinh Dang: Conceptualization; Methodology; Formal analysis; Software; Investigation; Validation; Visualization; Writing – original draft; Writing – review and editing.

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